



27

Handball

● ● ● ● ● ● ● ●
Martin Buchheit

Performance Demands of Handball

Every sport is unique in terms of how it's played, how it's won, and the associated demands. Handball is no exception. In this section, we introduce the sport and discuss the various performance factors for outfield players and the relative contribution that physical performance makes toward achieving team performance and winning.

Sport Description and Factors of Winning

Handball, also known as team handball or Olympic handball, is a team sport played by ~20 million players throughout the world involving two teams of seven players (six outfield players and a goal-keeper), which hand-pass a ball toward the aim of eventually throwing it into the other team's goal. A standard match consists of two periods of 30 min on a 40 × 20 m indoor court. The team that scores the most goals wins. Handball is referred to as a *transition game* because players frequently switch between defensive and offensive play, and the game is characterized by frequent high-intensity body contacts, along with intermittent running and sprinting.

Relative Contribution of Physical Performance

As for all ball-based team sports, technical skills, tactical awareness, and game intelligence are without doubt the main contributors of success. Players with limited ability to pass the ball precisely, shoot, or make the right decisions in the moment will never become champions, irrespective of the physical fitness attributes they may possess or develop. Fitness is, however, important to support the development of technical performance on the field. While physical fitness will never guarantee success, a lack of any given physical attribute can be the cause of failure (figure 27.1).

Targets of Physical Performance in Handball

Anthropometric characteristics, as well as high levels of strength, muscle power, and throwing velocity, are the most important generalized physical factors needed for gaining an advantage in handball that would translate into an improved probability of success at the elite level (21, 27). Additionally, as in most team sports, physical demands are clearly position dependent in terms of movement patterns, volume of high-intensity actions, and associated work:recovery



Figure 27.1 The position of the handball player on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation in handball, acknowledging this varies between outfield positions (24) except the goalkeeper, who is a different beast and does not require much high-intensity interval training. Outside of critical psychological and team chemistry components, which are difficult to quantify, the pie chart shows the general relative importance of skills, tactical awareness, and physical capacities for handball success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

ratios (24), which require further levels of individualization when it comes to conditioning players. As player demands are clearly position dependent, the following discussion describes the specific physical attributes and needs of the different player positions.

Outfield Players

In addition to the general physical targets, development of aerobic power should not be underestimated. During match play, players run from 4 to 6 km (24) at a mean intensity of roughly 80% to 90% of maximal heart rate (HR_{max}) (25). Significant associations between maximal oxygen uptake ($\dot{V}O_{2\max}$) and playing level have also been shown (21, 27). In fact, elite players have been shown to repeat more than 120 high-intensity actions during a game (24). Thus, a well-developed aerobic system is likely to be beneficial for metabolic recovery between the explosive efforts associated with successful handball match performance (29, 31).

Goalkeepers

For the goalkeepers, there is no need to generate excessive neuromuscular fatigue from running, so the better strategy is to remove them from the run

conditioning exercises and have them do some general aerobic conditioning on a stationary cycle ergometer or perform goalkeeper-specific training exercises.

Key Weapons, Manipulations, and Surveillance Tools

Recall that weapons refer to the high-intensity interval training (HIIT) formats we can use to target the physiological responses of importance, while the surveillance tools are what we are using to monitor the individual responses to those weapons. In this section, we present the different HIIT weapons, their manipulations, along with ways that we can monitor their effectiveness (surveillance).

HIIT Weapons

As outlined, there are five different metabolic and/or neuromuscular load targets I aim for throughout the season (see figure 1.5 (7)):

Type 1: metabolic O_2 system

Type 2: metabolic O_2 system + neuromuscular

Type 3: metabolic O_2 + anaerobic systems

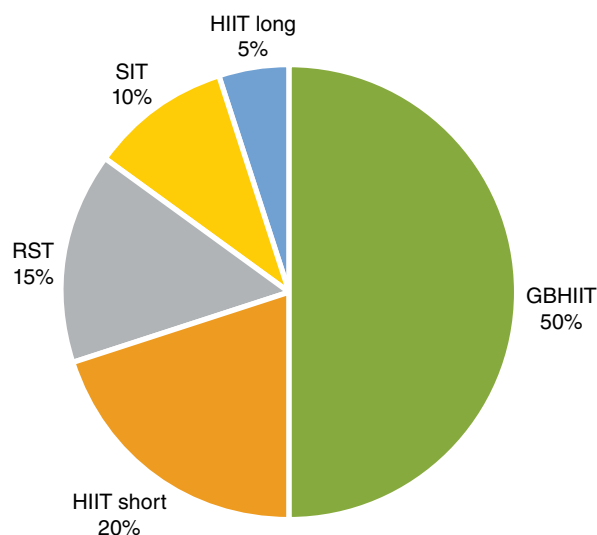


Figure 27.2 Percentage of the different HIIT weapons (formats) used throughout the annual season in elite handball. HIIT: high-intensity interval training; RST: repeated-sprint training; SIT: sprint interval training; GBHIIT: game-based HIIT.

Type 4: metabolic O_2 + anaerobic systems and neuromuscular

Type 5: metabolic anaerobic system and neuromuscular

The five HIIT weapons used to hit these targets were detailed earlier in chapters 4 and 5 (8). For handball (figure 27.2), the large majority of these are game-based HIIT, i.e., small-sided games (SSGs) (50%, essentially in-season), followed next by short intervals (20%, some preseason, but mostly in-season), repeated-sprint training (RST) (15%, in-season only), sprint interval training (SIT) (10%, in season only), and long intervals (5%, preseason exclusively).

Manipulations of Interval Training Variables

The running intensity for each HIIT weapon I've used has tended to be almost fixed for each format; rather, adjustments are achieved through programming variations between the different days/weeks. For example, if a short-interval HIIT weapon is programmed to target type 1 on a given day/week, RST, SIT, or SGGs formats might be used to target types 3 and 4 for the next day/week, which also adds variation to the training content.

HIIT With Long Intervals (Outdoor)

I generally implement HIIT with long intervals in the outdoor environment, either on a track, on a

soccer pitch (grass), or in the forest (with distances marked on the path), so as to minimize the musculoskeletal strain (7) (avoiding HIIT type 5, and possibly type 4 targets (7)). These typical long-interval weapons are generally performed over 3 to 4 min at 90% of $V_{Inc.Test}$ or 85% V_{IFT} . This represents 750 to 950 m efforts completed over 3 min, depending on a player's fitness, with athletes running 3 to 5 repetitions interspersed by 2 min of passive recovery. Players are generally spread across 4 or 5 groups (15 km/h, 16 km/h, 17 km/h, 18 km/h, and >18 km/h for $V_{Inc.Test}$ or 17 km/h, 18 km/h, 19 km/h, 20 km/h, and >21 km/h for V_{IFT}) and are requested to reach group-specific cones set across the running track at appropriate times. These sessions are generally prescribed at the end of the week, before a day off, so that players can recover from an anticipated neuromuscular fatigue (see the example in table 27.1 at the end of the chapter). Additionally, I try to schedule these at the end of the day so that athletes may benefit from a greater $\dot{V}O_2$ slow component, i.e., higher $\dot{V}O_2$ for a similar or lower running speed due to muscle fatigue and loss of metabolic efficiency (8).

HIIT With Short Intervals

My preferred HIIT short-interval weapon formats include 15 s on/15 s off, 20 s/20 s, and more often 10 s/20 s (see tables 27.1 to 27.4 at the end of the chapter), which can be used to target almost all HIIT types (1, 2, 3, and 4). I chose these formats as they can be associated with a limited level of neuromuscular fatigue and anaerobic contribution (especially 10 s/20 s, HIIT type 1), which helps preserve the quality of the conjoined handball sequences (same session) as well as the strength/speed sessions planned the following day (12, 13). We generally spread our players into 5 groups (17 km/h, 18 km/h, 19 km/h, 20 km/h, and >21 km/h for V_{IFT}), and we request that they run over group-based distances using cones on the court. For example, for players with a V_{IFT} of 19 km/h, and for a 15 s/15 s HIIT format run at 95% of V_{IFT} (relief interval: passive), the target distance will be $(19/3.6) \times 0.95 \times 15 = 75$ m (19 is divided by 3.6 to convert the speed from km/h to m/s, for convenience) (3). However, since most runs must be performed with changes of direction (CODs) to fit in the 40 m handball court, the time needed for COD must be considered when calculating the target run distance in order to ensure a similar cardiorespiratory load compared with straight-line runs, while remaining close to type 1 target loading (or in the worst-case scenario, type 2 but at the lower

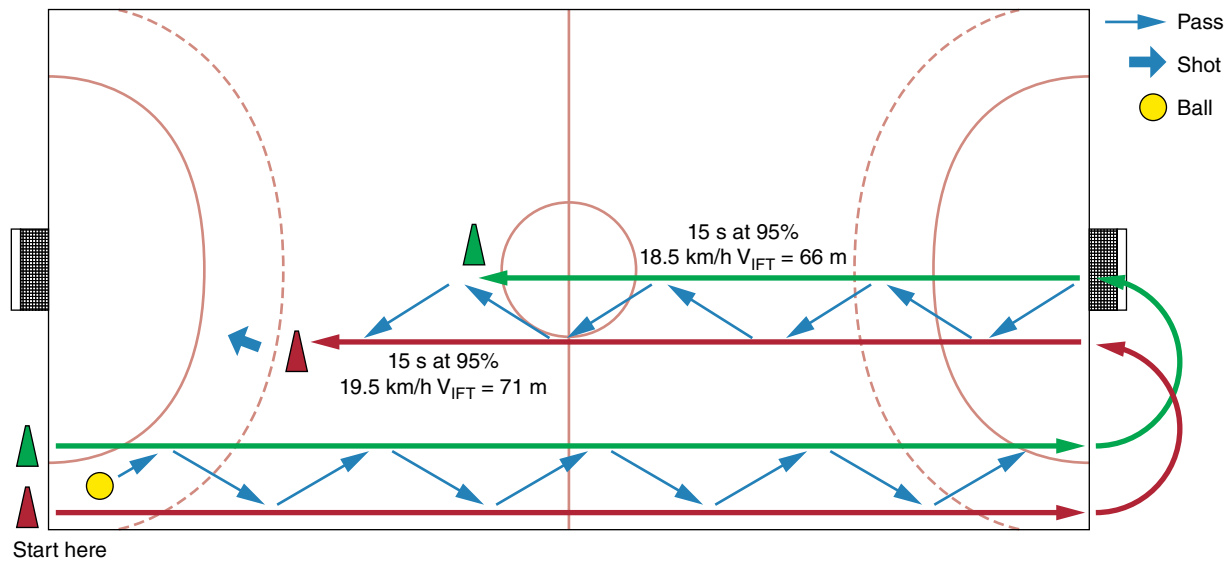


Figure 27.3 Example of handball-specific high-intensity interval training with short intervals (15 s/15 s format, type 3 target) based on V_{IFT} . The exercise consists of two players passing the ball continuously while running at a set speed and finishes with a shot on goal. The exercise is easier if players with similar V_{IFT} run together, but can also be implemented using individualized distances. In this example, the player on the red course is fitter than the player on the green course, so he must cover 5 m more within the same time period.

Reprinted by permission from M. Buchheit, "Programming High-Intensity Training in Handball," *Aspetar Sports Medicine Journal* 3, Targeted Topic: Sports Medicine in Handball (April 2014): 120-128.

end of neuromuscular constraint for this target type). Therefore, in relation to the estimated energetic cost of COD during HIIT (see chapter 2, V_{IFT} section), if the players have to run over a 40 m shuttle, they would instead cover 71 m. If the shuttle length is divided in half (i.e., a 20 m shuttle), the distance to cover drops to 65 m (3). A spreadsheet that completes this calculation for 15 players at a time is available from the author on request.

To make HIIT with short intervals more handball specific, we often integrate the ball into the activity on different occasions. For example, the players can run together and continuously pass the ball to one another, before the last player with the ball at the end shoots at the goal (figure 27.3). Another way to individualize HIIT with short intervals is to use the position-specific work:rest ratio and/or game-specific effort distribution (24). Under this situation, wingers need to perform HIIT with more intense runs interspersed by longer rest periods (for example, 10 s (110% of V_{IFT})/ 20 s (passive)), compared with backs (20 s (95%)/ 20 s (jog) or 30 s (90%)/ 30 s (jog)) (24).

Finally, merging specific defensive action maneuvers into short intervals is also possible; however, at least two important aspects must be considered. First,

repeating such actions for the entire duration of the work bout (i.e., 10 to 20 s) reduces game specificity. That is, in a game, a defender generally only reaches an attacker once or twice per defensive action before the game is either stopped (technical fault) or continues on across other players. Performing defensive actions throughout the 10 to 20 s period and accumulating a number of repetitions would amount to an inappropriate number of defensive maneuvers. Second, only using defensive movements does not allow control over the exercise interval intensity, as it does with run-based drills. Thus, an attractive option is to combine the defensive actions for 5 s with individualized running tasks occurring for the remaining 10 or 15 s of the interval (figure 27.4).

Repeated-Sprint Training

For the RST format (type 4 target), I generally implement 2 to 4 sets of 5 to 8 sprints across 15 to 30 m distances, interspersed with 14 to 25 s of passive or active (jog or ~45% V_{IFT}) recovery (7) during the first part of the training session. I intersperse these between two training sequences. For example, I might implement the following: warm-up, 10 to 20 min technical sequences, RST, prior to the

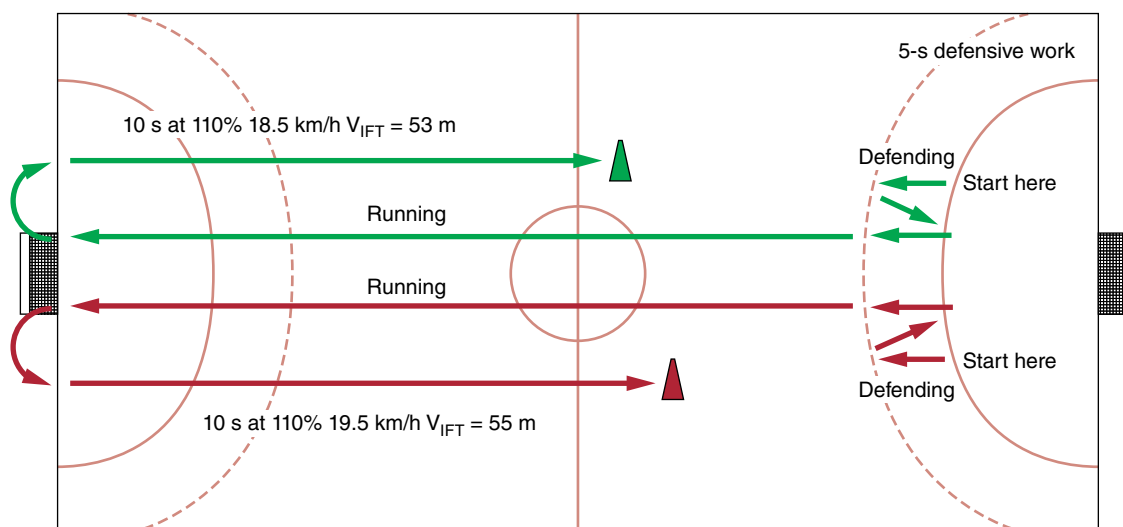


Figure 27.4 Defensive actions performed for 5 s are combined with an individualized running task for the remaining 10 s of the 15 s interval (short interval; type 3 and 4 targets). This allows control of the exercise intensity (running section) while preserving the specificity of the defensive task (players generally do not defend continuously for 15 to 20 s).

Reprinted by permission from M. Buchheit, "Programming High-Intensity Training in Handball," *Aspetar Sports Medicine Journal* 3, Targeted Topic: Sports Medicine in Handball (April 2014): 120-128.

remainder of the training day (12). See, for example, the video at <https://www.youtube.com/watch?v=nb4Ft9cyW6M>. The RST weapon is well suited for team relays and when it comes to implementing position-specific sequences, such as counterstrike work for wingers (figure 27.5). Sprints can also be performed in shuttle format, which would, from an injury-prevention perspective, lower stride length and subsequently hamstring strain risk (8). As with short-interval HIIT weapons, game demands of sprinting can be used to design position-specific repeated-sprint sequences (24), such as 20 to 30 m sprints and 10 to 15 m sprints for wingers and backs, respectively. The programming of these sequences (see table 27.4) should be well considered due to the associated very large anaerobic contribution and neuromuscular fatigue that may impact the quality of the strength, speed, and tactical work performed during subsequent sequences or sessions. It is also important to discontinue programming RST at the end of a demanding session, when fatigue may acutely increase injury risk.

Sprint Interval Training

I've used SIT weapons in the form of 3 to 6 all-out 30 s shuttle sprints over 40 m, interspersed with 2 to 4 min of passive recovery (type 5 target) (7, 10). Given the intensity and duration of the format, it is

difficult to involve ball work during these drills; however, it can nevertheless be added at the end of the runs. Due to the high anaerobic contribution and large neuromuscular fatigue associated with SIT, its programming during the sessions and during the week is similar to that discussed above for RST (see table 27.3).

Game-Based HIIT Training

As with most team sports, and described in depth in chapters 2 and 5, I tend to organize game-based training in the format of small-sided games (SSGs, figure 27.6) involving a varying number of players, generally less than during official competitions, <https://www.youtube.com/watch?v=26VKkeRDTMM>, over various field dimensions (e.g., 12 × 24 m, 30 × 15 m, 32 × 16 m, or 40 × 20 m, figures 27.7 and 27.8 (17)). In regional-to-national level players, we've shown that 4 versus 4 handball SSGs (+ 2 goalkeepers, 40 × 20 m, full handball court) are an effective means of achieving and maintaining a high percentage of $\dot{V}O_2\text{max}$ during training, despite lower blood lactate levels compared to run-based HIIT (14) (type 2 target). In addition, players tend to perceive SSGs as less painful (<https://www.youtube.com/watch?v=X8kuZ4i0B6A>).

In my experience, I have tended to simplify or modify typical handball rules with the aim of

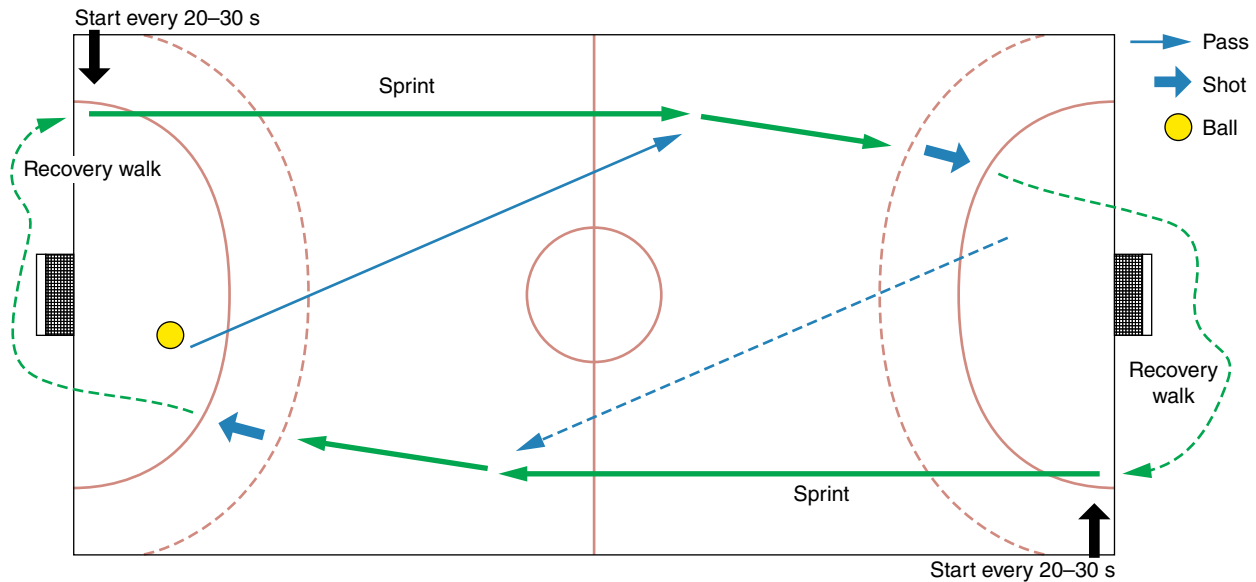


Figure 27.5 Example of a repeated-sprint sequence format (type 4 target) in which players sprint, receive the ball from the goalkeeper, and shoot before preparing to start again on the other side of the court (6 repetitions). With mean exercise time around 6 s per effort, up to 5 players can play at the same time (player rotation ~30 s).

Reprinted by permission from M. Buchheit, "Programming High-Intensity Training in Handball," *Aspetar Sports Medicine Journal* 3, Targeted Topic: Sports Medicine in Handball (April 2014): 120-128.

maximizing physiological responses for the following reasons.

To avoid game breaks that would unnecessarily reduce exercise intensity:

1. dribbling and defense contacts are not allowed; infringement of minor technical rules (such as walking) are not sanctioned;
2. throw-on after a goal is immediately made by goalkeepers from their 6 m area;
3. coaches are always available to immediately replace the ball when it is kicked away from the playing area; and
4. all four players must be in the opponent's half of the court for a goal to be validated (9).

To ensure that all players perform at the desired intensity, since exercise intensity cannot be controlled during SSGs—as with run-based drills—we must rely on player motivation to ensure that all players commit to the required (likely maximum) intensity during the drills. For that reason, encouragement and winning objectives between the teams are key. However, since players of varying fitness levels must play together (to preserve game-specific relationships between players), the relative work of

the fitter players can wind up being relatively lower than that of the less fit. Indeed, a negative correlation ($r = -0.88$) has been shown between relative exercise intensity during a 4 versus 4 SSG (% of $\dot{V}O_2\text{max}$ maintained during the SSG) and a player's $\dot{V}O_2\text{max}$ as measured during the 30-15 IFT (14). This suggests that, over time, fitter players might not benefit enough from these drills to improve their aerobic power/endurance capacity. To overcome these limitations, additional rules can be implemented, including:

- whenever an identified fitter player releases the ball, he performs two push-ups (which increases overall energy demands);
- whenever an identified fitter player releases the ball, he has to place one foot on the court's side-lines before coming back into the game (which increases his overall running demands);
- one identified fitter player is always involved in the attacking team, so that his volume of play is increased.

While there are infinite ways to modulate exercise intensity during a run-based HIIT (7), little data is available to guide training manipulation based on physiological loading during handball SSGs. Simple modification of rules, such as permission for player

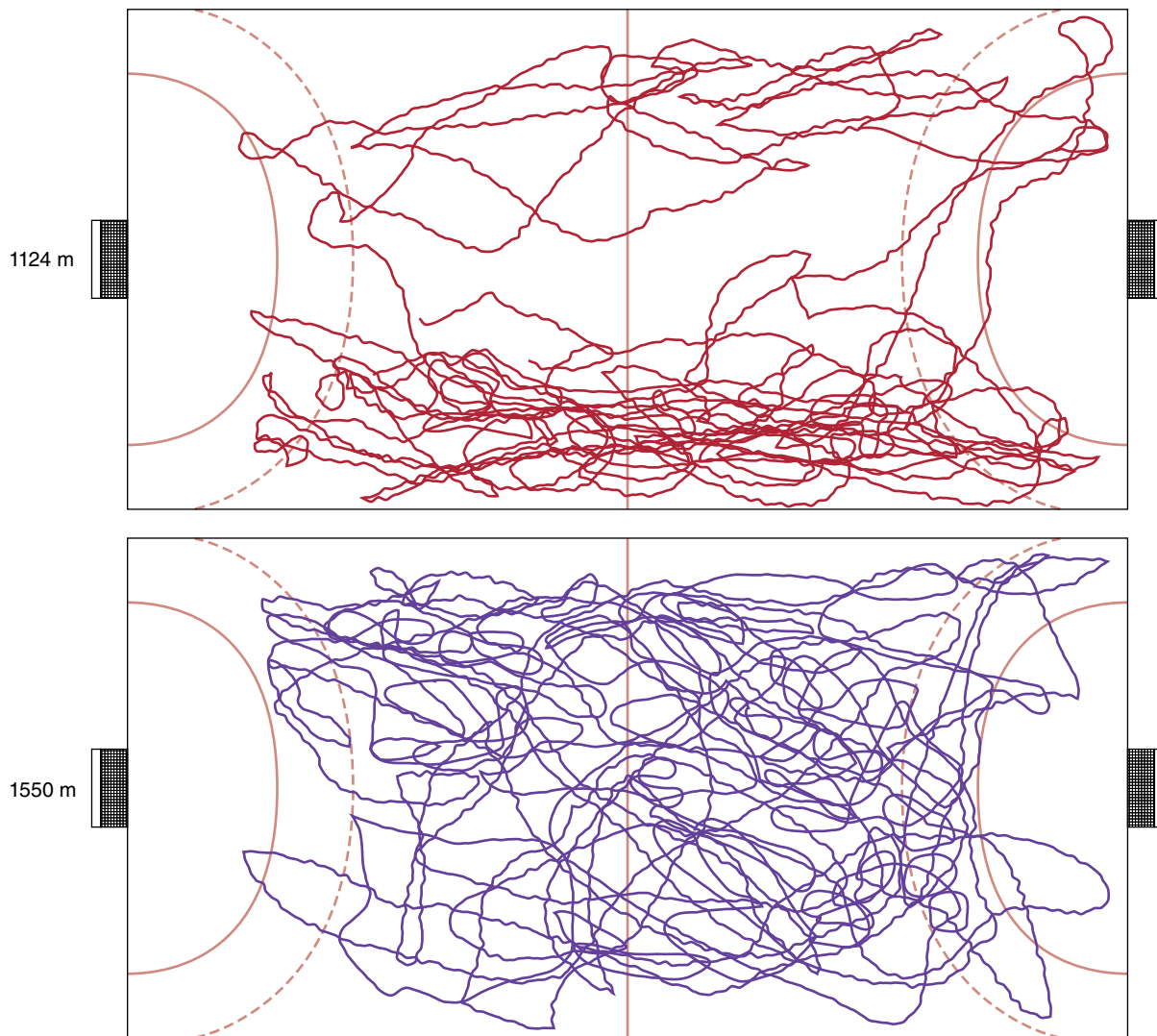


Figure 27.6 Distance covered by two different players (a, $V_{IFT} = 18.5$ km/h; b, $V_{IFT} = 20.5$ km/h) during an 8 min, 4 versus 4 small-sided handball game over the full court (12).

Reprinted by permission from M. Buchheit, "Programming High-Intensity Training in Handball," *Aspetar Sports Medicine Journal* 3, Targeted Topic: Sports Medicine in Handball (April 2014): 120-128.

contact or not, is probably the first alteration that I use in practice. For example, in national junior players, noncontact 3-a-side SSGs were shown to be associated with increased rating of perceived exertion (RPE) and a 50% greater sprinting distance despite similar HR responses than contact-based games (19). We also often alter court dimensions, as in other team sports, to manipulate the acute physiological demands of handball games (17).

Metabolic Responses to Changes in Court Dimensions An increase in playing area has shown no predictable effect on HR responses during SSGs (figure 27.7) (14, 17). This is likely related to the fact

that, over smaller playing areas, the number of changes in velocity, COD, and contacts increases, compared to larger pitches (17), which compensates for the lower running-only demands (14, 17). It is also worth noting that, because of the static phases and handball-specific muscular contractions, which increase HR independently of muscle O_2 demands, HR responses during SSGs tend to become dissociated from $\dot{V}O_2$ values. Therefore, extrapolating metabolic responses from HR is limited in this special situation (15).

Locomotor Responses to Changes in Court Dimensions In contrast to HR, we tend to see a linear relationship between running demands and

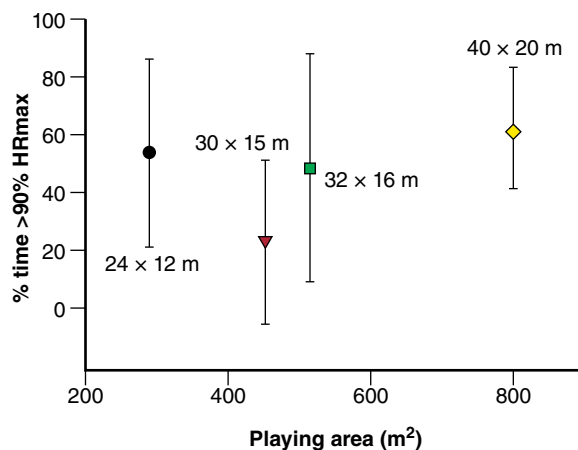


Figure 27.7 Time spent above 90% of maximal heart rate (HRmax, 90% confidence intervals) during 3 versus 3 small-sided games with different playing areas.

Reprinted by permission from M. Buchheit, "Programming High-Intensity Training in Handball," *Aspetar Sports Medicine Journal* 3, Targeted Topic: Sports Medicine in Handball (April 2014): 120-128.

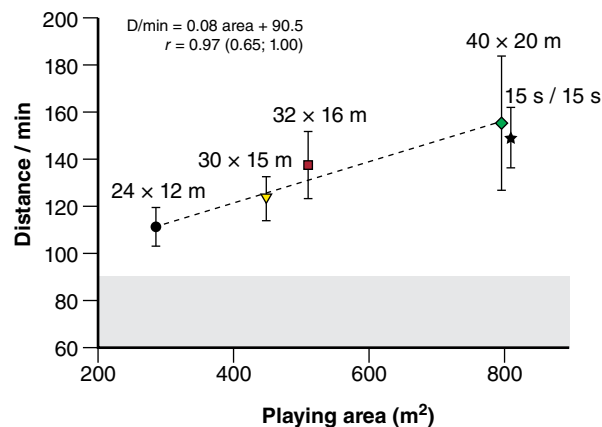


Figure 27.8 Distance covered per minute as a function of playing area during 3 versus 3 small-sided handball games (90% confidence intervals). The gray area represents distance per minute during games (24). The distance per minute covered during the short-interval exercise format (15 s/15 s) is also provided for comparison.

Reprinted by permission from M. Buchheit, "Programming High-Intensity Training in Handball," *Aspetar Sports Medicine Journal* 3, Targeted Topic: Sports Medicine in Handball (April 2014): 120-128.

playing area (figure 27.8). Additionally, the running pace reached during four different SSGs (17) has actually been shown to be higher than that during a match (24). Thus, SSGs represent an appropriate overload for the locomotor and likely the metabolic (peripheral at least) systems (figure 27.8). Interestingly,

playing 4 versus 4 on the full handball pitch allows players to reach the same running pace as during short-interval HIIT (figure 27.8), which is ~50% greater than game demands (see chapter 30 for a similar training approach).

In summary, changes in court dimensions or game rules can be used, as with other team sports, to manipulate the acute physiological demands of handball games, where the larger the playing area, the greater the running pace, but not necessarily the HR. Further research is needed to improve SSG programming, including research into best duration, repetitions, number of players, rules, and playing position-specific formats. If some specific rules are implemented to increase the relative exercise intensity of the fitter players, SSGs represent an interesting alternative to traditional HIIT, especially for in-season training content (see table 27.4).

The main advantages of SSGs include:

1. lower anaerobic contribution in training (type 2 targeting),
2. perceived as less painful by the majority of players,
3. time with the ball is maximized,
4. training interest and enjoyment is improved, and
5. the format allows work under fatigue to develop important handball components, such as agility, reaction time, and hand-eye coordination.

The only disadvantage of SSGs compared to run-based short-interval HIIT is the likely greater neuromuscular work that may impact the quality of the strength, speed, and tactical work performed during subsequent training sessions, but this would be very likely less than RSS and SIT by comparison.

Longitudinal Training Effects of HIIT

Figure 27.9 summarizes the effects of different HIIT interventions that I conducted in highly trained young players over several years in France in the late 2000s (National Talent Programs) (10, 12, 13). It shows that both high-intensity running capacity (V_{IFT}) and repeated-sprint performance (average time of six repeated sprints) can be improved during the season after 4 to 10 wk of HIIT supplementation (~2 sessions per wk, in addition to usual training) in highly trained players (~10 hours of training + 1 competitive game

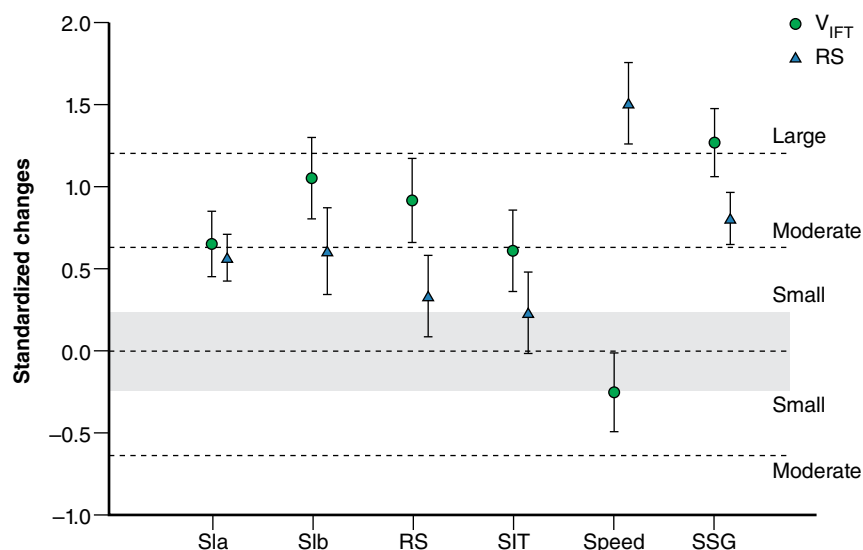


Figure 27.9 Standardized changes in the peak speed reached in the 30-15 intermittent fitness test (V_{IFT}) and mean repeated-sprint time following different training interventions based on short intervals (SI), repeated-sprint sequences (RS), sprint interval sessions (SIT), speed agility (Speed), and small-sided games (SSG) in highly trained young handball players. The gray area represents trivial changes. The short-interval (SIa) program consisted of 2 sessions per week involving 6 to 12 min of intermittent running for 15 s (95% V_{IFT}) interspersed with 15 s of passive recovery for 10 consecutive weeks (13). The short-interval (SIb) program consisted of 2 sessions per week involving 6 to 12 min of intermittent running for 15 s (95% V_{IFT}) to 20 s (90% V_{IFT}) interspersed with 15 to 20 s of passive recovery for 9 consecutive weeks (12). The repeated-sprint training program occurred across 9 consecutive weeks and consisted of 2 to 3 sets of 5 to 6 $\times$ 15 to 20 m shuttle sprints interspersed with 14 s of passive or 23 s of active recovery (45% V_{IFT}) (12). The SIT, conducted over 4 consecutive weeks, involved 3 to 6 repetitions of 30 s all-out shuttle sprints over 40 m, interspersed with 2 min of passive recovery (10). Speed training took place over 4 consecutive weeks and consisted of 4 to 6 series of 4 to 6 exercises (e.g., agility drills, standing start, and very short shuttle sprints, all of <5 s duration); each repetition and series was interspersed with at least 30 s and 3 min of passive recovery, respectively (10). SSG training consisted of small handball games (2 to 4 $\times$ 2 min 30 s to 4 min games) for 10 consecutive weeks (13).

Reprinted by permission from M. Buchheit, "Programming High-Intensity Training in Handball," *Aspetar Sports Medicine Journal* 3, Targeted Topic: Sports Medicine in Handball (April 2014): 120-128.

per week). Importantly, all changes reported in figure 27.9 (irrespective of the HIIT weapon format) can be considered substantial because they were all small to large in magnitude and clearly greater than their smallest worthwhile change (22) (the gray area represents trivial to nonsubstantial changes). Interestingly, the magnitude of improvement in both V_{IFT} ($r=0.56$) and mean repeated-sprint time ($r=-0.75$) after HIIT training (excluding speed) was linearly related to training duration (number of weeks). In addition, the magnitude of improvement in the average repeated-sprint time after HIIT training (excluding speed) was inversely related to baseline performance ($r=-0.68$); however, this relationship was unclear for V_{IFT} . The magnitude of improvement in mean repeated-sprint time (excluding speed) and

V_{IFT} was also largely correlated (-0.73). Therefore, while it is difficult to compare the respective effects of the different interventions because of the differences in training duration and baseline performance, the 10 wk SSG program was the only training method associated with large and moderate improvements in V_{IFT} and repeated-sprint performance, respectively (13). In all the aforementioned experimental training phases, only one type of HIIT was used (for study design purposes). Of course, the optimal physical conditioning scenario likely involves a combination of all types of HIIT supplementation (see table 27.5). Further studies comparing these HIIT programs over a similar duration in players matched for baseline physical performance are required to draw definitive conclusions.

Load Surveillance and Monitoring Tools

Despite some very recent developments of inertial sensors (26, 32) and semiautomatic video systems (16), the use of wearables and tracking technologies is still very limited in handball and is mainly reserved for research purposes. To our knowledge, a noninvasive practical tracking device has yet to be produced that can be used during training sessions, as might be implemented in other sports such as soccer or rugby. For this reason, despite its limitations (see above), I generally monitor HR during some training sessions, including the various HIIT formats, and collect RPEs after each session (see chapter 8). Training load can then be calculated via the popular session-RPE method (23).

Training Status Surveillance and Monitoring Tools

As described in chapter 9, the training status monitoring tools that I've used in handball are generally limited to noninvasive and low-cost measurements and include wellness questionnaires (overall well-being (28)), jumps (neuromuscular status (8)), exercise HR (fitness (6)), and resting heart rate variability (HRV, overall training status (6)). Since the early 2000s, we have been using HRV to track Sélestat (Senior French First League) player adaptation to intense preseason training cycles. Figure 27.10 shows how heterogeneous the training response can be between players exposed to the same training stimulus, but with large differences in volume (see story at the end of the chapter). We used this individual response to tailor each player's subsequent training. Additionally, we often use the HR and RPE responses to HIIT sequences as markers of adaptations (the lower, the better the player's fitness when comparing all similar sequences between players) (11).

Strategies for Structuring the Training Program Using HIIT

Strategies for structuring HIIT programming in handball may be confined by various controllable and uncontrollable factors before proper progression and periodization can be appropriately implemented. This section discusses some of the nuances of the training program and HIIT structuring in handball.

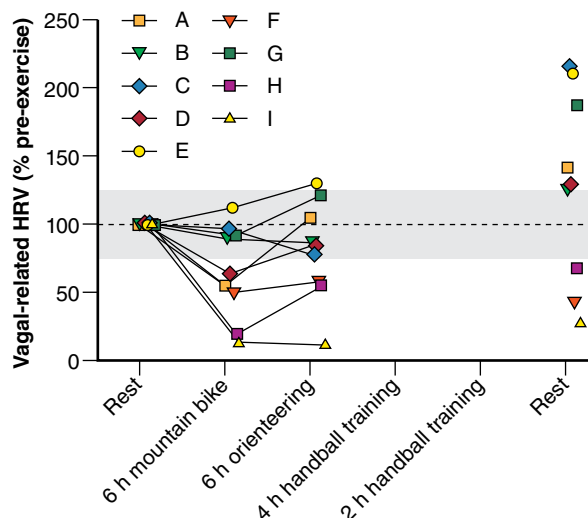


Figure 27.10 Individual cardiac parasympathetic responses of elite handball players during 4 consecutive days of training. The heart rate variability (HRV) values correspond to the measurement taken in the morning (5 min, supine at home) after a rest day, mountain biking, orienteering (running), or handball-specific training including SSGs. Shaded area represents the coefficient of variation for vagal-related HRV indices under resting conditions (1).

Reprinted by permission of Springer Nature from J. Stanley, J.M. Peake, and M. Buchheit, "Cardiac Parasympathetic Reactivation Following Exercise: Implications for Training Prescription," *Sports Medicine* 43, no. 12 (2013): 1259-1277.

Controllable and Uncontrollable Factors

In my experience, the only truly uncontrollable factors within my training program design are the match locations and game schedule timing, which are controlled by the sport federation. Minor to this is the constant requirement for gym-based work (strength and power) combined with high-quality technical sessions, which together constrain the timing, volume, and objective of the HIIT sequences within a given training week. Thus, the chosen HIIT session must induce the least possible subsequent level of fatigue. In fact, I believe that residual neuromuscular fatigue post-HIIT can reduce a player's attention and technical proficiency during tactical sessions, as well as his force production capacity and rate of force application during subsequent (strength/speed) sessions, which likely impairs the learning process and attenuates the training stimuli needed for optimal neuromuscular adaptation (2, 20).

In-season, when match schedules dictate the training programming, there are likely to be only two weekly scenarios, with 1 or 2 matches per week. In the latter case, there winds up being no room for HIIT. In contrast, I generally program 1 or 2 HIIT sequences in the week when we have a full week between matches (see table 27.4).

Program Structure and Progression

In handball, as with many team sports, there actually is very little periodization that occurs, except during the preseason (see table 27.5). In-season, I try to keep the weekly training load almost constant throughout the months and focus on the successive match preparation-match-recovery cycles, with a short emphasis on development during midweek days when possible (18). For these reasons, the skeleton of the weekly program (see table 27.4) and the program objectives remains similar across the season.

Incorporation of HIIT

In the off-season, I often program heavy-loading run-based HIIT sessions using long intervals after handball sessions (and not during), mainly due to the fact that these need to be implemented outdoors

(forest, field, track). These can be achieved in the off-season relative to the in-season, as the level of exhaustion created is substantial and affects subsequent training.

In-season, the HIIT sequences become fully incorporated into technical handball sessions over 1 or 2 d in the middle of the week (see table 27.4), and as such are performed in between technical and tactical drills during the main session. The objective of these low-volume, mainly handball-specific HIIT sequences is mostly maintenance of cardiorespiratory fitness, while limiting anaerobic contribution, neuromuscular stress, and fatigue before subsequent day gym sessions. It is worth noting, however, that in some scenarios, like substitute players who lack competitive time, a high neuromuscular load during HIIT might be desired. Thus, neuromuscular-demanding HIIT (e.g., RST) may help such players prepare best for specific match demands (24).

Sample Training Programs

The following tables provide examples of HIIT programming during different phases of training, i.e., preseason (tables 27.1 to 27.3) and in-season (table 27.4). Note the distribution of the different HIIT types in relation to the microcycle's objectives.

Table 27.1 $\dot{V}O_2$ Development Cycle (First Week[s] of Preseason; Typically, 1 to 2 Week Duration) (8)

Microcycle (physical emphasis)		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
$\dot{V}O_2$ development++	AM	Rest	Handball	Generic strength/ injury prevention	Handball	Rest	Handball	Rest
	PM	Handball Type 3 HIIT with long intervals (grass) 5×3 min (85% $V_{Inc.Test}$)/ 2 min (0) or type 3 HIIT with short intervals (grass) 2×12 min 30 s (85% V_{IFT})/ 30 s (40% V_{IFT})	Handball	Handball	Type 1 HIIT with short intervals (grass) 2×12 min 15 s (95% V_{IFT})/ 15 s (40% V_{IFT})	Handball	Type 4 HIIT with long intervals (soft outdoor track) 5×3 min (90% $V_{Inc.}$ $Test$)/ 2 min (0) HIIT	

Note that 2 d are allowed after the lactic session (30 s/30 s) and that the long-bout HIIT (greater speed and performed on a slightly harder outdoor surface) is programmed prior to the day off. Type refers to the physiological target.

Table 27.2 $\dot{V}O_2$ and Strength/Speed Development Cycle
(Generic Phase, Preseason; Typically 2 Weeks Duration) (8)

Microcycle (physical emphasis)		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
$\dot{V}O_2$ + strength/ speed develop- ment (generic)	AM	Generic strength/ speed/ injury prevention	Rest	Generic power/speed	Rest	Generic strength/ speed	Handball Type 3 HIIT with short intervals (grass) 2 × 10 min 20 s (95% V_{IFT})/20 s (0)	Rest
	PM	Handball	Handball Type 1 HIIT with short intervals (grass) 2 × 10 min 10 s (110% V_{IFT})/20 s (0)	Handball	Handball	Handball	Rest	

Note the de-emphasis of lactic work in Tuesday's HIIT session. Both HIIT sessions have limited neuromuscular demands to allow leg freshness for the strength/speed development. Type refers to the physiological target.

Table 27.3 $\dot{V}O_2$ and Strength/Speed Development Cycle
(Specific Phase, Preseason; Typically 2 Weeks Duration) (8)

Microcycle (physical emphasis)		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
$\dot{V}O_2$ + strength/ speed development (specific)	AM	Specific strength/ speed/ injury prevention	Rest	Type 2 GBHIIT 2 × 3 min (SSG) 4 vs. 4 RPE 6 + Type 1 6 min 10 s (100% V_{IFT} shuttle)/10 s (0)	Rest	Specific power/ speed	Handball	Rest
	PM	Handball	Handball	Handball	Handball	Rest	Handball Type 5 SIT 6 × 30 s (all-out shuttle)/4 min (0)	

Note that the lactic session is programmed before the day off. Type refers to the physiological target.

Table 27.4 Maintenance (In-Season; Remainder of the Season, Typically 9 to 10 Months) (8)

Microcycle (physical emphasis)		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Competitive (in season)	AM	Recovery/ strength/ injury prevention	Rest	Rest	Strength/ power/ speed	Rest	Rest	Rest
	PM	Handball	Handball Type 1 HIIT with short intervals 1 × 10 min 10 s (110% V_{IFT} shuttle)/20 s (0) For substitutes or if the power/speed session of Thursday is canceled: Type 4 2 × RST 6 × 20 m-90° COD/25 s (0) with 15 min (hand- ball) between RSS	Handball Type 2 GBHIIT 3 × 4 min (SSG) 4 vs. 4 RPE 7	Handball	Handball	Game	

Note that the RST session is performed with 90° COD to reduce glycolytic anaerobic energy release and is performed only by substitutes or when the speed/power session is missed (to partially compensate for the neuromuscular work at high intensity). SSGs can also be selected for their specificity, to lower the anaerobic glycolytic energy contribution 3 d before the game. Type refers to the physiological target.

CASE STUDY

I was privileged to be in the position charged with preparing the First French League Handball Team of Sélestat in 2004-2005 for a season that would become one of the best in its history. It was my second season at the club, and I had gained the benefit of my experience from the first year, along with knowing the players and staff better, knowing their individual needs, and gaining their trust. Additionally, through some trial and error, I had developed a better understanding of what worked and what didn't; that is, I knew the training stimulus (load) that was triggering adaptation for certain players versus others. Together with Francois Berthier, the head coach, our strategy was to increase the aerobic conditioning of the team using a greater training volume in the first 2 weeks. Our plan involved using a challenging 4 d camp in the mountains to permit some team integration for a few new players who had joined the squad over the summer. The training involved some epic mountain biking tours as well as a short 6 h mountaineering session in the hilly forest of the Voges, not to mention evening handball practice alongside a few run-based HIITs.

The rules of the mountaineering race were relatively simple: players were given a map when they woke (without breakfast) along with the key checkpoints that had to be reached to obtain their food and drink for the day; the winning team needed to reach all locations in the minimum amount of time. To be sure the players provided their best effort and were motivated, the volume of HIIT applied in the remainder of the week (table 27.5) was to be inversely related to the mountaineering results shown from the eight different pairs of players. The first team would do what was planned, the second team would perform an additional 3 min effort, and so forth.

On the morning of the race, the faces of certain players said it all. A social night at the hotel got carried away, creating the unfortunate state we were seeing in front of us (hangovers; little sleep). In return for this, the coaching staff opted to lengthen the orienteering course out a ways to provide a statement and indicate intent for the year. At the same time, unbeknownst to us, a few team leaders had already in fact moved checkpoints of their fellow teammates to challenge them further.

At the end of the day, the players' return times ranged from late in the afternoon to very late at night. Nevertheless, the deep levels of fatigue, mental and physical challenge, along with the necessary cooperation to achieve the tasks caused strong team bonds to be formed that I feel brought us together as a team from very early on. From that point, however, I told the players that their attitude to our camp and subsequent consequence had compromised my plan, and that I wasn't sure how to best adapt their training load in the following days. Some had been in the forest for 12 h, while others for only 4. The predicament could not have played into my hands any better, and I was able to then suggest to the team that I could try using HRV to adjust their training load accordingly. Desperate that the suggestion might work in their favor, the guys bought in, and you can view the data from this story in figure 27.10 (notice the massive effect in some players from the mountaineering event). Importantly, most guys continued to make the effort to take the morning measurement throughout the remainder of the year.

The preseason went on as planned, with successful friendly games and no major injuries. At the end of 6 wk, the boys were doing fine on the court, but many were complaining of heavy legs. While there is no direct link between a player's physical capacities and game outcomes, the team had the worst start to the season in years. After many stressful nights, chats with colleagues and mentors and comparisons with past year training plans, I attributed the reported heavy legs to an overreliance on RST during the last 3 wk of the preparation. After a few weeks into the season, I changed the programming toward handball-specific but isolated speed and HIIT training sessions (short interval and SSGs, table 27.4) (4), and neuromuscular freshness returned, in line with noticeable team freshness and success. We went on to make some big wins and made the quarterfinals of the French Cup, losing in the match by a few points. Nevertheless, it was a year in my sporting life that as a practitioner

Table 27.5 Focus on the HIIT Prescription of the 6 Week Preseason Conditioning Program Used in July 2004 With Sélestat Handball Team (First French League)

Type refers to the physiological target of the HIIT format.

	HIIT	Repeated-sprint training	Sprint interval training	Small-sided games	Total HIIT time (T at VO ₂ max)
Week 1	(1) Short Type 1 2×6 min 15 s [95%] / 15 s [jog] + (2) Short Type 3 2×6 min 20 s [95%] / 20 s [passive]				~24 min (15-18 min)
Week 2	(1) Short Type 3 2×6 min 30 s [90%] / 30 s [jog] + (2) Long Type 4 4×3 min [80%] / 2 min [passive]			(3) Type 2 3×3 min 4 vs. 4	~33 min (18-21 min)
Week 3	(1) Short Type 3 10 min 15 s [100%] / 15 s [passive] + (2) Long Type 4 5×3 min [80%] / 2 min [passive]			(3) Type 2 3×4 min 4 vs. 4	~37 min (20-24 min)
Week 4	(1) Short Type 1 6 min 10 s [95%] / 10 s [passive]		(2) Type 5 6×30 s every 2 min (40 m shuttles)	(3) Type 2 4×4 min 4 vs. 4	~35 min (12-15 min)
Week 5		(1) Type 4 3×(6×15+15 m shuttle every 30 s) (2) Type 4 3×(4×20+20 m shuttle every 30 s)		(3) Type 2 3×3 min 4 vs. 4	~26 min (10-12 min)
Week 6	(1) Short Type 1 6 min 10 s [105%] / 20 s [passive]	(1) Type 4 2×(4×15+15 m shuttle every 30 s)		(2) Type 2 3×4 min 4 vs. 4	~22 min (10-14 min)

The numbers in parentheses indicate the number of separate days when the different HIIT formats were implemented. The numbers in brackets refer to running intensity as either a percentage of V_{IF} or recovery jog. Number of players for small-sided games refers to field players. Note the distribution of HIIT-type targets. For example, when a type 4 target is introduced week 2, a type 3 target is replaced by a type 2 target to compensate for the overall anaerobic contribution. When a type 5 target is introduced week 4, the other session targets are limited to type 1 and 2 to limit weekly neuromuscular and anaerobic participation. Note also the overreliance on RST the last 3 weeks (as discussed in the text).

(continued)

CASE STUDY *(CONTINUED)*

I was proud to contribute to. I made mistakes, which drove me harder toward understanding the reasons for them, and I learned and gained experience as a result. As a team, we came together to achieve something special for a little club with the league's second smallest budget. The experience showed me the importance of training monitoring (HRV) to better manage (sometimes unplanned) variations in load, as well as how to program and solve the training program puzzle to use the different HIIT formats in relation to an ever-evolving context.